

International Institute of Information Technology

HS0.302- Research Methods in Human Sciences

Spring 2026

Mid-Semester Exam

Time: 90 minutes

Marks: 30

1. a) Explain social engineering as a tenet of positivism.
b) How is positivist thinking on social sciences expressed in Popper's falsifiability criterion?
c) What is the fundamental difference between exploratory and confirmatory data analysis and which one do you think is more important in human sciences?

4+6+5

2. Answer the following questions with clear explanations based on the cross-country regression results tabulated in page 2. The independent variables are listed at the left of the table. The regression coefficients for the independent variables in models (1) to (6) are presented in respective columns of the table. Assume the critical value of the t-statistic at 5% level of significance for a two-tailed test with the relevant degrees of freedom is 2.00. Read the notes below the table carefully.
 - a) Using suitable notations write down the underlying population relationship that is estimated in model (3)
 - b) What is the computed t value for the variable $\ln(\text{exports/area})$ in model (3)?
 - c) What is the interpretation of the coefficient of $\ln(\text{exports/area})$ in model (3)?
 - d) What is the interpretation of the coefficient for French legal origin (Base = Others) in model (4)?
 - e) Which model has the best explanatory power in the table and why? Answer using statistical as well as intuitive reasoning.
 - f) Can any of the models in the table be used to claim a causal relationship between slave exports in colonial times and per capita income? Why or why not?

2 marks each



RELATIONSHIP BETWEEN SLAVE EXPORTS AND INCOME

Dependent variable is log real per capita GDP in 2000, $\ln y$

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{exports/area})$	-0.112*** (0.024)	-0.076*** (0.029)	-0.108*** (0.037)	-0.085** (0.035)	-0.103*** (0.034)	-0.128*** (0.034)
Distance from equator		0.016 (0.017)	-0.005 (0.020)	0.019 (0.018)	0.023 (0.017)	0.006 (0.017)
Longitude		0.001 (0.005)	-0.007 (0.006)	-0.004 (0.006)	-0.004 (0.005)	-0.009 (0.006)
Lowest monthly rainfall		-0.001 (0.007)	0.008 (0.008)	0.0001 (0.007)	-0.001 (0.006)	-0.002 (0.008)
Avg max humidity		0.009 (0.012)	0.008 (0.012)	0.009 (0.012)	0.015 (0.011)	0.013 (0.010)
Avg min temperature		-0.019 (0.028)	-0.039 (0.028)	-0.005 (0.027)	-0.015 (0.026)	-0.037 (0.025)
$\ln(\text{coastline/area})$		0.085** (0.039)	0.092** (0.042)	0.095** (0.042)	0.082** (0.040)	0.083** (0.037)
Island indicator				-0.398 (0.529)	-0.150 (0.516)	
Percent Islamic				-0.008*** (0.003)	-0.006* (0.003)	-0.003 (0.003)
French legal origin				0.755 (0.503)	0.643 (0.470)	-0.141 (0.734)
North Africa indicator				0.382 (0.484)	-0.304 (0.517)	
$\ln(\text{gold prod/pop})$					0.011 (0.017)	0.014 (0.015)
$\ln(\text{oil prod/pop})$					0.078*** (0.027)	0.088*** (0.025)
$\ln(\text{diamond prod/pop})$					-0.039 (0.043)	-0.048 (0.041)
Colonizer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number obs.	52	52	42	52	52	42
R^2	.51	.60	.63	.71	.77	.80

Notes. OLS estimates of (1) are reported. The dependent variable is the natural log of real per capita GDP in 2000, $\ln y$. The slave export variable $\ln(\text{exports/area})$ is the natural log of the total number of slaves exported from each country between 1400 and 1900 in the four slave trades normalized by land area. The colonizer fixed effects are indicator variables for the identity of the colonizer at the time of independence. Coefficients are reported with standard errors in brackets. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

3. Answer the following questions using Table IV (below) which presents two stage least square (2SLS) results for the same model analysed in the previous table. Assume the critical value of the t-statistic at 5% level of significance for a two-tailed test with the relevant degrees of freedom is 2.00.
- What is the rationale of conducting a 2SLS estimation for the model?
 - Which variable(s) have been used as instrumental variable(s)?
 - Are the instruments used in the model valid? Give reasons.

2+1+2

TABLE IV
ESTIMATES OF THE RELATIONSHIP BETWEEN SLAVE EXPORTS AND INCOME

	(1)	(2)	(3)	(4)
Second Stage. Dependent variable is log income in 2000, $\ln y$				
$\ln(\text{exports/area})$	-0.208*** (0.053) [-0.51, -0.14]	-0.201*** (0.047) [-0.42, -0.13]	-0.286* (0.153) [-∞, +∞]	-0.248*** (0.071) [-0.62, -0.12]
Colonizer fixed effects	No	Yes	Yes	Yes
Geography controls	No	No	Yes	Yes
Restricted sample	No	No	No	Yes
F-stat	15.4	4.32	1.73	2.17
Number of obs.	52	52	52	42
First Stage. Dependent variable is slave exports, $\ln(\text{exports/area})$				
Atlantic distance	-1.31*** (0.357)	-1.74*** (0.425)	-1.32* (0.761)	-1.69** (0.680)
Indian distance	-1.10*** (0.380)	-1.43*** (0.531)	-1.08 (0.697)	-1.57* (0.801)
Saharan distance	-2.43*** (0.823)	-3.00*** (1.05)	-1.14 (1.59)	-4.08** (1.55)
Red Sea distance	-0.002 (0.710)	-0.152 (0.813)	-1.22 (1.82)	2.13 (2.40)
F-stat	4.55	2.38	1.82	4.01
Colonizer fixed effects	No	Yes	Yes	Yes
Geography controls	No	No	Yes	Yes
Restricted sample	No	No	No	Yes
Hausman test	.02	.01	.02	.04
(p-value)				
Sargan test (p-value)	.18	.30	.65	.51

Notes. IV estimates of (1) are reported. Slave exports $\ln(\text{exports/area})$ is the natural log of the total number of slaves exported from each country between 1400 and 1900 in the four slave trades normalized by land area. The colonizer fixed effects are indicator variables for the identity of the colonizer at the time of independence. Coefficients are reported, with standard errors in brackets. For the endogenous variable $\ln(\text{exports/area})$, I also report 95% confidence regions based on Moreira's (2003) conditional likelihood ratio (CLR) approach. These are reported in square brackets. The p-value of the Hausman test is for the Wu-Hausman chi-squared test. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. The "restricted sample" excludes island and North African countries. The "geography controls" are distance from equator, longitude, lowest monthly rainfall, avg max humidity, avg min temperature, and $\ln(\text{coastline/area})$.